🎯 MISMATHE

RT-02 Mock Test — Full Syllabus Replica

MHT-CET Pattern · Prep for RT-02 (Repeat) · Sun 2 Aug 2026 · Class 11

⏱️ Duration: 3 hours (9:00 AM – 12:00 PM) | 📝 150 Questions · 200 Marks | ✅ No negative marking | 📘 Physics 50 · Chemistry 50 · Maths 50

INSTRUCTIONS

- This paper covers exactly RT-02's syllabus: **Physics** — Units and Measurements, Mathematical Methods; **Chemistry** — Some Basic Concepts of Chemistry, Introduction to Analytical Chemistry; **Maths** — Trigonometry-I, Trigonometry-II.
- Each Physics and Chemistry question carries **1 mark**. Each Maths question carries **2 marks**.
- There is **no negative marking** — attempt every question.
- Answer key with one-line workings is at the end. Don't peek until you've finished a full section.

Section	Chapters	Questions	Marks
Physics	Units & Measurements + Mathematical Methods	50	50
Chemistry	Some Basic Concepts + Intro to Analytical Chemistry	50	50
Mathematics	Trigonometry-I + Trigonometry-II	50	100
Total		**150**	**200**

PHYSICS — Q1–50

1 mark each

UNITS AND MEASUREMENTS

1. How many fundamental (base) physical quantities are there in the SI system?

A) 5 **B)** 6 **C)** 7 **D)** 8

2. The SI unit of luminous intensity is the:

A) lumen **B)** candela **C)** lux **D)** watt

3. Which of the following is a derived unit?

A) metre **B)** kilogram **C)** newton **D)** second

4. The dimensional formula of force is:

A) MLT^-1 **B)** MLT^-2 **C)** ML^2T^-2 **D)** ML^-1T^-2

5. The dimensional formula of work is:

A) MLT^-2 **B)** ML^2T^-2 **C)** ML^2T^-1 **D)** ML^-1T^-2

6. The number of significant figures in 0.00420 is:

A) 2 **B)** 3 **C)** 5 **D)** 6

7. Accuracy of a measurement refers to:

A) how close repeated readings are to each other
B) how close the measured value is to the true value
C) the number of decimal places recorded
D) the least count of the instrument used

8. Precision of a measurement refers to:

A) closeness of the measured value to the true value
B) closeness among repeated measurements of the same quantity
C) the smallest quantity an instrument can measure
D) complete absence of error

9. The order of magnitude of the number 8500 is:

A) 10^2 **B)** 10^3 **C)** 10^4 **D)** 10^5

10. The absolute error in a measurement is defined as:

A) the magnitude of the difference between the mean value and an individual measured value
B) the ratio of the error to the true value
C) the percentage deviation from the mean
D) the least count of the instrument

11. The product of 12.5 and 2.5, expressed to the correct number of significant figures, is:

A) 31.250
B) 31.2
C) 31
D) 30

12. The sum of 4.2 and 3.146, expressed to the correct number of significant figures, is:

A) 7.3
B) 7.35
C) 7.346
D) 7.4

13. Rounding off 3.14659 to three significant figures gives:

A) 3.14
B) 3.15
C) 3.146
D) 3.1

14. If the percentage error in measuring the radius of a sphere is 2%, the percentage error in its calculated volume is:

A) 2%
B) 4%
C) 6%
D) 8%

15. A physical quantity X is given by X = a^2 b^3 / c. If the percentage errors in a, b, c are 1%, 2%, 3% respectively, the percentage error in X is:

A) 6%
B) 9%
C) 11%
D) 13%

16. A resistance is measured as 100 ohm with an absolute error of 2 ohm. The relative (fractional) error in the measurement is:

A) 0.01
B) 0.02
C) 0.2
D) 2

17. Dimensional analysis can be used to check the correctness of an equation, but it fails when the equation involves:

A) sums of terms having the same dimensions
B) purely numerical (dimensionless) constants
C) products of physical quantities
D) powers of physical quantities

18. One newton is equal to how many dyne?

A) 10^2 dyne
B) 10^3 dyne
C) 10^5 dyne
D) 10^7 dyne

19. Five readings of the length of a rod are 2.63, 2.56, 2.42, 2.71 and 2.48 cm. The mean absolute error in the measurement is nearest to:

A) 0.09 cm
B) 0.07 cm
C) 0.14 cm
D) 0.44 cm

20. The dimensional formula of the universal gravitational constant G is:

A) M^-1L^3T^-2
B) ML^3T^-2
C) M^-1L^2T^-2
D) ML^-3T^2

21. In the formula g = 4 x pi^2 x L / T^2, if the percentage errors in measuring L and T are 2% and 1% respectively, the percentage error in g is:

A) 3%
B) 4%
C) 5%
D) 6%

22. The velocity v of a particle depends on time t as v = at + b/(t+c), where a, b, c are constants. The dimensional formula of b is:

A) LT^-1
B) LT^-2
C) L
D) T

23. The universal gas constant R has the value 8.31 J/(mol K) in SI units. Its value in CGS units, i.e. erg/(mol K), is:

A) 8.31x10^5
B) 8.31x10^7
C) 8.31x10^9
D) 8.31x10^-7

24. A physical quantity P is given by P = a^2 b^3 / (c^0.5 x d). The percentage errors in a, b, c, d are 1%, 2%, 3%, 4% respectively. The percentage error in P is:

A) 11%
B) 12.5%
C) 13.5%
D) 15%

25. The density of a cube-shaped block is calculated from its mass and the length of its side. If the mass is measured with 3% error and each side with 1% error, the maximum percentage error in the calculated density is:

A) 4%
B) 5%
C) 6%
D) 9%

MATHEMATICAL METHODS

26. According to the laws of logarithms, log(mn) is equal to:

A) log m - log n
B) log m + log n
C) log m x log n
D) n log m

27. **According to the laws of logarithms, log(m/n) is equal to:**

A) log m + log n
B) log m x log n
C) m log n
D) log m - log n

28. **d/dx(x^5) = ?**

A) x^4
B) 5x^6
C) 5x^4
D) 4x^4

29. **d/dx(sin x) = ?**

A) cos x
B) -cos x
C) -sin x
D) sec^2 x

30. **d/dx(e^x) = ?**

A) x e^(x-1)
B) e^x/x
C) 1
D) e^x

31. **d/dx(log x) = ? (natural log)**

A) x
B) 1/x
C) log x
D) 1/x^2

32. **integral of x^n dx (n not equal to -1) = ?**

A) n x^(n-1) + C
B) x^(n-1)/(n-1) + C
C) x^(n+1)/(n+1) + C
D) x^n/n + C

33. **integral of cos x dx = ?**

A) sin x + C
B) -sin x + C
C) -cos x + C
D) cos x + C

34. **integral of e^x dx = ?**

A) e^x/x + C
B) x e^x + C
C) log(e^x) + C
D) e^x + C

35. **For small x, the binomial approximation (1+x)^n is approximately equal to:**

A) 1 + x^n
B) 1 + nx
C) n + x
D) 1 + n x^2

36. **Using the product rule, d/dx(x^2 sin x) = ?**

A) x^2 sin x + 2x cos x
B) x^2 cos x - 2x sin x
C) x^2 cos x + 2x sin x
D) 2x cos x - x^2 sin x

37. **Using the quotient rule, d/dx(sin x / x) = ?**

A) (x cos x - sin x)/x^2
B) (sin x - x cos x)/x^2
C) (x cos x + sin x)/x^2
D) cos x / x

38. **d/dx(sin 2x) = ?**

A) cos 2x
B) -2 cos 2x
C) 2 sin 2x
D) 2 cos 2x

39. **The position of a particle is x = 3t^2 + 2t + 5 (x in m, t in s). Its velocity at t = 2 s is:**

A) 12 m/s
B) 14 m/s
C) 16 m/s
D) 8 m/s

40. **The slope of the tangent to the curve y = x^2 at the point x = 3 is:**

A) 9
B) 3
C) 6
D) 12

41. **The area under the curve y = x between x = 0 and x = 2 is given by the definite integral (0 to 2) of x dx. Its value is:**

A) 2
B) 4
C) 1
D) 8

42. **A particle moves with velocity v = 4t (m/s). Using the area under the v-t graph, its displacement between t = 0 and t = 3 s is:**

A) 12 m
B) 24 m
C) 9 m
D) 18 m

43. **A body is projected vertically upward with velocity u. Using h = ut - (1/2)g t^2, the time taken to return to the point of projection (h = 0, t not equal to 0) is:**

A) u/g
B) 2u/g
C) u/(2g)
D) g/u

44. **Using the binomial approximation (1+x)^n approx 1 + nx, the approximate value of (0.99)^4 is:**

A) 0.94
B) 1.04
C) 0.96
D) 0.9996

45. **If log x - log 5 = log 4, then x = ?**

- **A)** 20
- **B)** 9
- **C)** 0.8
- **D)** 45

46. **If y = (sin 3x)^2, then dy/dx = ?**

- **A)** 3 sin 3x cos 3x
- **B)** 2 sin 3x cos 3x
- **C)** 6 sin 3x
- **D)** 6 sin 3x cos 3x

47. **The value of the definite integral (0 to 1) of 2x e^(x^2) dx is:**

- **A)** e
- **B)** e - 1
- **C)** e + 1
- **D)** 1 - e

48. **The displacement of a particle is s = t^3 - 6t^2 + 9t (s in m, t in s). The particle is momentarily at rest at t = ?**

- **A)** 1 s and 3 s
- **B)** 0 s and 3 s
- **C)** 2 s and 4 s
- **D)** 1 s and 2 s

49. **A stone is projected vertically upward with velocity 30 m/s (g = 10 m/s^2). Using h = ut - (1/2)g t^2, the times at which the stone is at a height of 25 m are:**

- **A)** 2 s and 4 s
- **B)** 0.5 s and 4.5 s
- **C)** 1 s and 5 s
- **D)** 1 s and 4 s

50. **In a circuit, the power delivered is P = EI - I^2 R, where E and R are constants. The current I for which power P is maximum is:**

- **A)** E/R
- **B)** 2E/R
- **C)** R/(2E)
- **D)** E/(2R)

CHEMISTRY — Q51–100

1 mark each

SOME BASIC CONCEPTS OF CHEMISTRY

51. According to the law of conservation of mass, in a chemical reaction the total mass of the reactants

A) is always greater than the total mass of the products
B) is always less than the total mass of the products
C) remains equal to the total mass of the products
D) has no fixed relationship with the mass of the products

52. Two samples of pure water, one collected from a river and another prepared in the laboratory, both contain hydrogen and oxygen combined in the same mass ratio of 1:8. This observation illustrates the

A) law of conservation of mass
B) law of multiple proportions
C) law of definite proportions
D) Gay-Lussac's law

53. Carbon forms two oxides, CO and CO_2. For a fixed mass of carbon, the masses of oxygen combining in these two oxides are in the ratio 1:2. This is an example of the

A) law of conservation of mass
B) law of definite proportions
C) law of multiple proportions
D) law of reciprocal proportions

54. One volume of nitrogen combines with three volumes of hydrogen to give two volumes of ammonia, all volumes measured at the same temperature and pressure. This illustrates

A) law of definite proportions
B) Gay-Lussac's law of gaseous volumes
C) law of conservation of mass
D) Dalton's atomic theory

55. Avogadro's law states that equal volumes of all gases, under the same conditions of temperature and pressure, contain

A) equal masses
B) equal number of molecules
C) equal number of atoms
D) equal number of protons

56. Which of the following is NOT a postulate of Dalton's atomic theory?

A) Atoms are indivisible particles which can neither be created nor destroyed
B) Atoms of the same element are identical in mass and properties
C) Atoms of different elements combine in simple whole number ratios to form compounds
D) Atoms can be divided into protons, neutrons and electrons

57. Boron consists of two isotopes, boron-10 (mass number 10, 20% abundance) and boron-11 (mass number 11, 80% abundance). The average atomic mass of boron is approximately

A) 10.2
B) 10.5
C) 10.8
D) 11.0

58. What is the molar mass of sulphuric acid, H_2SO_4? (Atomic masses: H=1, S=32, O=16)

A) 82 g/mol
B) 98 g/mol
C) 100 g/mol
D) 114 g/mol

59. How many moles are present in 4.4 g of CO_2? (Molar mass of CO_2 = 44 g/mol)

A) 0.1 mol
B) 0.2 mol
C) 1.0 mol
D) 4.4 mol

60. The number of molecules present in 0.5 mole of O_2 gas is ($N_A = 6.022\times10^{23}$ per mol)

A) 6.022×10^{23}
B) 3.011×10^{23}
C) 1.2044×10^{24}
D) 3.011×10^{22}

61. The volume occupied by 0.25 mole of an ideal gas at STP is (molar volume at STP = 22.4 L/mol)

A) 2.24 L
B) 5.6 L
C) 11.2 L
D) 22.4 L

62. A compound contains 40% carbon, 6.7% hydrogen and 53.3% oxygen by mass. Its empirical formula is (Atomic masses: C=12, H=1, O=16)

A) CHO
B) CH_2O
C) $C_2H_4O_2$
D) CH_3O

63. The empirical formula of a compound is CH_2O and its molar mass is 180 g/mol. Its molecular formula is (Atomic masses: C=12, H=1, O=16)

A) $C_2H_4O_2$
B) $C_3H_6O_3$
C) $C_6H_{12}O_6$
D) $C_5H_{10}O_5$

64. The percentage of oxygen by mass in calcium carbonate, $CaCO_3$, is (Atomic masses: Ca=40, C=12, O=16)

A) 12%
B) 40%
C) 48%
D) 60%

65. On complete decomposition, 50 g of $CaCO_3$ produces how much CaO, given $CaCO_3 \rightarrow CaO + CO_2$? (Molar masses: $CaCO_3$ = 100 g/mol, CaO = 56 g/mol)

A) 22 g
B) 28 g
C) 44 g
D) 56 g

66. For the reaction $N_2 + 3H_2 \rightarrow 2NH_3$, 4 mol of N_2 is mixed with 6 mol of H_2. The limiting reagent is

A) N_2
B) H_2
C) both are completely consumed
D) NH_3

67. 13 g of zinc is reacted with 7.3 g of HCl as per $Zn + 2HCl \rightarrow ZnCl_2 + H_2$. What mass of the reactant remains unreacted? (Zn=65, HCl=36.5)

A) 3.25 g Zn
B) 6.5 g Zn
C) 3.65 g HCl
D) 7.3 g HCl

68. The molarity of a solution containing 4 g of NaOH dissolved to make 500 mL of solution is (Molar mass NaOH = 40 g/mol)

A) 0.1 M
B) 0.2 M
C) 0.4 M
D) 2.0 M

69. 18 g of glucose ($C_6H_{12}O_6$, molar mass = 180 g/mol) is dissolved in 500 g of water. The molality of the solution is

A) 0.1 m
B) 0.2 m
C) 0.5 m
D) 1.0 m

70. A solution is prepared by dissolving 2 mol of glucose in 8 mol of water. The mole fraction of glucose in the solution is

A) 0.1
B) 0.2
C) 0.25
D) 0.8

71. The equivalent weight of H_2SO_4 (molar mass 98 g/mol, basicity 2) is

A) 24.5 g/eq
B) 49 g/eq
C) 98 g/eq
D) 196 g/eq

72. A sample of water contains 5 mg of dissolved lead per kg of water. The concentration of lead in ppm is

A) 0.5 ppm
B) 5 ppm
C) 50 ppm
D) 500 ppm

73. The density of a 2 M aqueous NaOH solution is 1.28 g/mL. Calculate its molality. (Molar mass of NaOH = 40 g/mol)

A) 1.35 m
B) 1.56 m
C) 1.67 m
D) 2.00 m

74. An organic compound contains carbon and hydrogen only. Complete combustion of a sample gives 0.66 g of CO_2 and 0.36 g of H_2O. What is the empirical formula of the compound?

A) CH_2
B) C_2H_5
C) C_3H_8
D) C_4H_{10}

75. 6 g of H_2 gas is mixed with 32 g of O_2 gas and allowed to react as $2H_2 + O_2 \rightarrow 2H_2O$. What is the maximum mass of water formed?

A) 18 g
B) 27 g
C) 36 g
D) 54 g

INTRODUCTION TO ANALYTICAL CHEMISTRY

76. Analytical chemistry is the branch of chemistry that deals with:

A) The synthesis of new compounds only
B) The qualitative and quantitative composition of substances
C) The rates and mechanisms of chemical reactions
D) The physical properties of matter only

77. Qualitative analysis is used to determine:

A) The exact amount of each constituent present in a sample
B) Which chemical constituents (elements, ions, or compounds) are present in a sample
C) The rate of a chemical reaction
D) The molecular structure of an unknown compound only

78. Quantitative analysis is used to determine:

A) Only the identity of constituents present in a sample
B) The amount or concentration of a known constituent present in a sample
C) The color of a compound
D) The melting point of a substance

79. Volumetric (titrimetric) analysis and gravimetric analysis are examples of:

A) Instrumental methods of analysis
B) Classical (chemical) methods of analysis
C) Spectroscopic methods of analysis
D) Chromatographic methods of analysis

80. Which of the following is classified as an instrumental method of analysis?

A) Titrimetric analysis
B) Gravimetric analysis
C) Colorimetric analysis
D) Precipitation analysis

81. Which of the following is generally regarded as the first step in a chemical analysis?

A) Measurement of the desired property
B) Collection of a representative sample
C) Calculation of the final result
D) Interpretation and reporting of results

82. Errors that have a definite cause, occur in a definite pattern, and can be minimized by careful technique are known as:

A) Indeterminate errors
B) Determinate (systematic) errors
C) Absolute errors
D) Relative errors

83. Errors that arise from small, uncontrollable variations, causing results to be randomly scattered above and below the true value, are called:

A) Determinate errors
B) Method errors
C) Indeterminate (random) errors
D) Instrumental errors

84. The term that describes how close an experimentally measured value is to the true or accepted value is:

A) Precision
B) Accuracy
C) Reliability
D) Resolution

85. In titrimetric analysis, the solution of accurately known concentration that is added gradually from a burette is called the:

A) Titrate
B) Titrant
C) Indicator
D) Analyte

86. Precision refers to:

A) Agreement of a measured value with the true value
B) Degree of agreement among a series of repeated measurements of the same quantity
C) The number of significant figures in a measurement
D) The sensitivity of an instrument

87. A burette that has not been properly calibrated delivers volumes that are consistently 0.05 mL less than the true volume in every titration performed with it. This type of error is classified as:

A) Personal error
B) Method error
C) Instrumental error
D) Indeterminate error

88. In a gravimetric estimation, a small amount of the precipitate remains soluble in the mother liquor and is lost during filtration, causing consistently low results. This is an example of:

A) Instrumental error
B) Method error
C) Personal error
D) Random error

89. An analyst who is slightly color-blind consistently overshoots the end point while using a color-change indicator in every titration. This is an example of:

A) Instrumental error
B) Method error
C) Personal error
D) Indeterminate error

90. The true value of the mass of a sample is 12.50 g. A student weighs it and records 12.42 g. The absolute error in the measurement is:

A) +0.08 g
B) -0.08 g
C) +0.92 g
D) -0.92 g

91. A volume of solution is known to be exactly 50.00 mL, but a measurement gives 49.50 mL. What is the percentage error in the measurement?

A) 0.5%
B) 1.0%
C) 1.5%
D) 5.0%

92. In a set of five replicate titre values, the measure of precision obtained by subtracting the lowest value from the highest value is called the:

A) Mean deviation
B) Standard deviation
C) Range
D) Percentage error

93. The number of significant figures in the measurement 0.02030 g is:

A) 2
B) 3
C) 4
D) 5

94. Compared to the range, the standard deviation is considered a better measure of precision because it:

A) Considers only the two extreme values in a data set
B) Takes into account the deviation of every individual measurement from the mean
C) Ignores all but the largest deviation
D) Cannot be calculated for more than two measurements

95. For a substance to be used as a primary standard in volumetric analysis, it must:

A) Be hygroscopic and readily absorb moisture from air
B) Have high purity, stable composition, and a known high equivalent weight
C) Decompose readily on storage
D) React slowly and incompletely with the titrant

96. **Sodium hydroxide (NaOH) cannot be used as a primary standard because it is:**

A) Insoluble in water

B) Hygroscopic and readily absorbs moisture and carbon dioxide from air, making its exact composition uncertain

C) Too expensive to prepare in pure form

D) Unreactive with acids

97. **In an acid-base titration using phenolphthalein, the end point is where the indicator visibly changes color, while the equivalence point is where the acid and base have reacted in exact stoichiometric proportion. These two points:**

A) Are always exactly identical for every titration

B) May differ slightly, and the difference between them is called the titration error

C) Are unrelated concepts with no connection

D) Occur only in redox titrations, not acid-base titrations

98. **20 mL of a H2SO4 solution of unknown normality exactly neutralizes 24 mL of 0.15 N NaOH solution. The normality of the H2SO4 solution is:**

A) 0.12 N

B) 0.15 N

C) 0.18 N

D) 0.20 N

99. **A 0.530 g sample of impure Na2CO3 is dissolved and titrated with 0.2 N HCl using methyl orange indicator; 40 mL of the acid is required for complete reaction. (Equivalent mass of Na2CO3 = 53) The percentage of Na2CO3 in the sample is:**

A) 40%

B) 53%

C) 80%

D) 100%

100. **While reading the burette meniscus, a student always positions his eye above the liquid level instead of at eye level. As a result, every titre value he records is smaller than the true value by nearly the same amount, though the values are very close to one another on repetition. This situation best illustrates:**

A) High precision and high accuracy, due to negligible random error

B) High precision but low accuracy, due to a determinate (personal) error

C) Low precision but high accuracy, due to random error

D) Low precision and low accuracy, due to instrumental error

MATHEMATICS — Q101–150

2 marks each

TRIGONOMETRY - I

101. sin(30deg) + cos(60deg) = ?

A) 0
B) 1/2
C) 1
D) 3/2

102. tan(45deg) x cos(0deg) = ?

A) 1
B) 0
C) 2
D) -1

103. cosec(30deg) = ?

A) 1
B) 1/2
C) root2
D) 2

104. sec(60deg) = ?

A) 1
B) 2
C) 1/2
D) root3

105. tan(60deg) / tan(30deg) = ?

A) 1
B) root3
C) 3
D) 1/3

106. sin(120deg) = ?

A) 1/2
B) -1/2
C) -root3/2
D) root3/2

107. cos(210deg) = ?

A) root3/2
B) -root3/2
C) 1/2
D) -1/2

108. tan(315deg) = ?

A) -1
B) 1
C) root3
D) -root3

109. sin^2(30deg) + cos^2(30deg) = ?

A) 0
B) 1/2
C) 1
D) 2

110. If sin(theta) is positive and cos(theta) is positive, theta lies in which quadrant?

A) I
B) II
C) III
D) IV

111. The period of sin(theta) and cos(theta) is:

A) 90deg
B) 180deg
C) 270deg
D) 360deg

112. The domain of tan(theta) excludes:

A) multiples of 180deg
B) odd multiples of 90deg
C) multiples of 360deg
D) multiples of 45deg

113. The range of sec(theta) is:

A) between -1 and 1 inclusive
B) all real numbers
C) less than or equal to -1, or greater than or equal to 1
D) greater than 0

114. If theta lies in the third quadrant, which statement is correct?

A) sin(theta) is positive and cos(theta) is positive
B) sin(theta) is positive and tan(theta) is negative
C) cos(theta) is positive and sin(theta) is negative
D) tan(theta) is positive and cos(theta) is negative

115. If sin(theta) is negative and tan(theta) is positive, theta lies in which quadrant?

A) III
B) I
C) II
D) IV

116. sin(-theta) x cos(-theta) = ?

A) sin(theta) cos(theta)
B) -sin(theta) cos(theta)
C) cos^2(theta)
D) -cos^2(theta)

117. sin(90+theta) = ?

A) sin(theta)
B) -sin(theta)
C) cos(theta)
D) -cos(theta)

118. tan(270-theta) = ?

A) tan(theta)
B) -tan(theta)
C) -cot(theta)
D) cot(theta)

119. sec^2(theta) - tan^2(theta) = ?

A) 1
B) 0
C) -1
D) 2

120. cosec^2(theta) - cot^2(theta) = ?

A) 0
B) 1
C) -1
D) 2

121. If sin(theta) + cosec(theta) = 2, then sin^2(theta) + cosec^2(theta) = ?

A) 4
B) 1
C) 2
D) 0

122. (1 + tan^2(theta))(1 - sin(theta))(1 + sin(theta)) = ?

A) sin^2(theta)
B) cos^2(theta)
C) 0
D) 1

123. If sec(theta) + tan(theta) = p, then sec(theta) - tan(theta) equals:

A) 1/p
B) p
C) -p
D) -1/p

124. tan(theta)/(1+sec(theta)) + (1+sec(theta))/tan(theta) = ?

A) cosec(theta)
B) 2cosec(theta)
C) 2sec(theta)
D) 2sin(theta)

125. If cos(theta) = -3/5 and theta lies in the third quadrant, then tan(theta) = ?

A) -4/3
B) 3/4
C) 4/3
D) -3/4

TRIGONOMETRY - II

126. If sinA = 3/5 and cosB = 12/13, where A and B are acute angles, find sin(A+B).

A) 16/65
B) 56/65
C) 63/65
D) 20/65

127. If cosA = 4/5 and sinB = 5/13, where A and B are acute angles, find cos(A-B).

A) 33/65
B) 56/65
C) 63/65
D) 48/65

128. Find the value of sin75deg.

A) (root6+root2)/4
B) (root6-root2)/4
C) (root3+root2)/4
D) (root2-root6)/4

129. If tanA = 1/2 and tanB = 1/3, find tan(A+B).

A) 5/6
B) 5/7
C) 6/5
D) 1

130. If cosA = 3/5, find cos2A.

A) 7/25
B) -7/25
C) 18/25
D) -1/25

131. If sinA = 3/5 and A is acute, find sin2A.

A) 7/25
B) 12/25
C) 24/25
D) -24/25

132. If tanA = 1/2, find tan2A.

A) 4/3
B) 1
C) 3/4
D) 4/5

133. Find the value of tan15deg.

A) 2+root3
B) root3-2
C) root3+1
D) 2-root3

134. If sinA = 5/13 and cosB = 3/5, where A and B are acute angles, find sin(A-B).

A) 33/65
B) -33/65
C) 56/65
D) 63/65

135. If cosA = 7/25 and A is an acute angle (first quadrant), find sin(A/2).

A) 4/5
B) 3/4
C) 3/5
D) 7/25

136. Express sin4x + sin2x as a product.

A) 2 sin3x cosx
B) 2 cos3x sinx
C) 2 sin3x sinx
D) 2 cos3x cosx

137. Express cos5x - cos3x as a product.

A) 2 sin4x sinx
B) -2 cos4x sinx
C) -2 sin4x cosx
D) -2 sin4x sinx

138. Express 2cos3x cosx as a sum.

A) cos4x - cos2x
B) cos4x + cos2x
C) sin4x + sin2x
D) cos2x - cos4x

139. If sinA = 1/3, find sin3A.

A) -23/27
B) 25/27
C) 23/27
D) 19/27

140. If cosA = 1/2, find cos3A.

A) -1
B) 1
C) 0
D) -1/2

141. In triangle ABC, a = 10, angle A = 30deg, angle B = 45deg. Find side b using the sine rule.

A) 5root2
B) 10
C) 20
D) 10root2

142. In triangle ABC, a = 5, b = 6, c = 7. Find cosA using the cosine rule.

A) 19/35
B) 5/7
C) 1/5
D) 3/7

143. In triangle ABC, the projection formula for side a is:

A) a = b cosB + c cosC
B) a = b sinC + c sinB
C) a = b cosC + c cosB
D) a = c cosC - b cosB

144. Simplify sin50deg + sin10deg.

A) cos20deg
B) 2cos20deg
C) sin20deg
D) cos40deg

145. Find the general solution of sinx = 1/2.

A) x = npi + pi/6
B) x = 2npi plusminus pi/6
C) x = npi plusminus pi/6
D) x = npi + (-1)^n (pi/6)

146. Find the general solution of cosx = -1/2.

A) x = 2npi plusminus pi/3
B) x = 2npi plusminus 2pi/3
C) x = npi plusminus 2pi/3
D) x = 2npi + (-1)^n (2pi/3)

147. Find the general solution of 2cos^2x - 3cosx + 1 = 0.

A) x = 2npi plusminus pi/3 only
B) x = npi or x = npi plusminus pi/3
C) x = 2npi or x = 2npi plusminus pi/3
D) x = 2npi or x = npi plusminus pi/3

148. Simplify (sin5x - sin3x)/(cos5x + cos3x).

A) tanx
B) cotx
C) tan4x
D) cot4x

149. Find the value of sin3A/sinA - cos3A/cosA.

A) 0
B) 1
C) sin2A
D) 2

150. In triangle ABC, a = root3+1, b = root3-1, and angle C = 60deg. Find side c.

A) 2root2
B) root6
C) root10
D) 2root3

ANSWER KEY + ONE-LINE WORKING — PHYSICS (Q1–50)

Q		Working
1	C	SI has 7 base quantities: length, mass, time, current, temperature, amount of substance, luminous intensity.
2	B	Candela (cd) is the SI base unit of luminous intensity.
3	C	Newton equals kg m/s^2, built from base units; the others are base units themselves.
4	B	Force = mass x acceleration = M x LT^-2 = MLT^-2.
5	B	Work = force x displacement = MLT^-2 x L = ML^2T^-2.
6	B	Leading zeros are not significant; the digits 4, 2, 0 are significant, giving 3.
7	B	Accuracy means closeness of a measured value to the true (actual) value.
8	B	Precision indicates reproducibility, i.e. how close repeated readings are to one another.
9	C	8500 = 8.5x10^3; since 8.5 is greater than 5, it rounds up to 10^4.
10	A	Absolute error is the magnitude of (mean value minus individual measured value).
11	C	12.5x2.5=31.25; result limited to 2 sig figs (least among the factors) gives 31.
12	A	Sum = 7.346; addition rule keeps the least decimal places (1, from 4.2), giving 7.3.
13	B	The digit after the third significant figure is 6 (5 or more), so 3.14 rounds up to 3.15.
14	C	Volume is proportional to r^3, so percentage error in V = 3 x 2% = 6%.
15	C	Percentage error = 2(1%) + 3(2%) + 1(3%) = 2+6+3 = 11%.
16	B	Relative error = absolute error divided by measured value = 2/100 = 0.02.
17	B	Dimensional analysis cannot detect errors in dimensionless numerical constants such as 1/2 or pi.
18	C	1 N = 1 kg m/s^2 = 10^3 g x 10^2 cm/s^2 = 10^5 g cm/s^2 = 10^5 dyne.
19	A	Mean = 2.56 cm; sum of absolute deviations = 0.44 cm; mean absolute error = 0.44/5 = 0.088, about 0.09 cm.
20	A	From F=Gm1m2/r^2, G=Fr^2/(m1m2)=(MLT^-2)(L^2)/M^2=M^-1L^3T^-2.
21	B	g is proportional to L/T^2, so percentage error = 2% + 2(1%) = 4%.
22	C	(t+c) must have dimension T; since b/(t+c) equals velocity, b = velocity x T = LT^-1 x T = L.
23	B	1 J = 10^7 erg, so R = 8.31x10^7 erg/(mol K).
24	C	Percentage error = 2(1)+3(2)+0.5(3)+1(4) = 2+6+1.5+4 = 13.5%.
25	C	Density = mass/side^3, so percentage error = 3% + 3(1%) = 6%.
26	B	Log of a product equals sum of logs: log(mn) = log m + log n.
27	D	Log of a quotient equals difference of logs: log(m/n) = log m - log n.
28	C	By power rule, d/dx(x^n) = n x^(n-1), so d/dx(x^5) = 5x^4.
29	A	Standard derivative: d/dx(sin x) = cos x.
30	D	The exponential function e^x is its own derivative.
31	B	Standard derivative: d/dx(log x) = 1/x.
32	C	Power rule for integration: integral of x^n dx = x^(n+1)/(n+1) + C.
33	A	Standard integral: integral of cos x dx = sin x + C.
34	D	Standard integral: integral of e^x dx = e^x + C.
35	B	Neglecting higher powers of small x in the binomial expansion gives (1+x)^n approx 1 + nx.
36	C	d/dx(x^2 sin x) = x^2(cos x) + sin x(2x) = x^2 cos x + 2x sin x.
37	A	(f/g)' = (f'g - fg')/g^2 = (x cos x - sin x)/x^2, with f = sin x, g = x.
38	D	By chain rule, d/dx(sin 2x) = cos 2x times d/dx(2x) = 2 cos 2x.
39	B	v = dx/dt = 6t + 2; at t = 2, v = 12 + 2 = 14 m/s.
40	C	dy/dx = 2x; at x = 3, slope = 2(3) = 6.
41	A	integral (0 to 2) x dx = [x^2/2] from 0 to 2 = 4/2 - 0 = 2.
42	D	s = integral (0 to 3) 4t dt = [2t^2] from 0 to 3 = 18 m.
43	B	h = 0 gives t(u - gt/2) = 0, so t = 0 or t = 2u/g; the non-zero root is 2u/g.
44	C	Write 0.99 = 1 + (-0.01); then (0.99)^4 approx 1 + 4(-0.01) = 0.96.
45	A	log(x/5) = log 4 gives x/5 = 4, so x = 20.

46	**D**	By chain rule, dy/dx = 2(sin 3x)(cos 3x)(3) = 6 sin 3x cos 3x.
47	**B**	Put u = x^2, du = 2x dx; integral becomes (0 to 1) of e^u du = e - 1.
48	**A**	v = ds/dt = 3t^2 - 12t + 9 = 3(t-1)(t-3) = 0, giving t = 1 s, 3 s.
49	**C**	30t - 5t^2 = 25 gives t^2 - 6t + 5 = 0, so (t-1)(t-5) = 0, t = 1 s, 5 s.
50	**D**	dP/dI = E - 2IR = 0 gives I = E/(2R); second derivative -2R less than zero confirms maximum.

ANSWER KEY + ONE-LINE WORKING — CHEMISTRY (Q51–100)

Q		Working
51	C	Mass is neither created nor destroyed in a chemical reaction (Lavoisier's law).
52	C	A given compound from any source always has the same constituent mass ratio.
53	C	For fixed mass of carbon, oxygen masses (16 g and 32 g) are in simple ratio 1:2.
54	B	N2 + 3H2 -> 2NH3 reacts in simple whole-number volume ratio 1:3:2.
55	B	Equal volumes of gases at same T and P contain equal number of molecules.
56	D	Dalton considered atoms indivisible; subatomic particles were discovered later.
57	C	Average mass = (10x0.20) + (11x0.80) = 2.0 + 8.8 = 10.8.
58	B	M(H2SO4) = 2(1) + 32 + 4(16) = 2+32+64 = 98 g/mol.
59	A	Moles = mass/molar mass = 4.4/44 = 0.1 mol.
60	B	Molecules = moles x NA = 0.5 x 6.022x10^23 = 3.011x10^23.
61	B	Volume = moles x 22.4 = 0.25 x 22.4 = 5.6 L.
62	B	Moles C:H:O = 40/12 : 6.7/1 : 53.3/16 = 3.33:6.7:3.33 = 1:2:1, so CH2O.
63	C	Empirical mass CH2O = 30; n = 180/30 = 6; molecular formula = (CH2O)6 = C6H12O6.
64	C	M(CaCO3) = 100; mass of O = 48; %O = (48/100)x100 = 48%.
65	B	Moles CaCO3 = 50/100 = 0.5; moles CaO = 0.5; mass CaO = 0.5x56 = 28 g.
66	B	4 mol N2 needs 12 mol H2, but only 6 mol H2 is available, so H2 is limiting.
67	B	HCl (0.2 mol) is limiting; consumes 0.1 mol Zn (6.5 g); Zn remaining = 13-6.5 = 6.5 g.
68	B	Moles NaOH = 4/40 = 0.1; Molarity = 0.1 mol/0.5 L = 0.2 M.
69	B	Moles glucose = 18/180 = 0.1; Molality = 0.1 mol/0.5 kg = 0.2 m.
70	B	Mole fraction = 2/(2+8) = 0.2.
71	B	Equivalent weight = molar mass/n-factor = 98/2 = 49 g/eq.
72	B	5 mg per kg = 5 mg per 10^6 mg = 5 parts per million.
73	C	1 L solution mass = 1280 g; NaOH = 80 g; water = 1200 g = 1.2 kg; molality = 2/1.2 = 1.67 m.
74	C	Moles C = 0.66/44 = 0.015; moles H = 2x0.36/18 = 0.04; ratio C:H = 1:2.67 = 3:8, so C3H8.
75	C	O2 (1 mol) is limiting; it consumes 2 mol H2 and produces 2 mol H2O = 36 g.
76	B	Analytical chemistry covers both identifying and measuring components of matter.
77	B	Qualitative analysis identifies which components are present, not their amount.
78	B	Quantitative analysis measures the amount or concentration of a known constituent.
79	B	Gravimetric and volumetric analysis are classical (wet chemical) methods.
80	C	Colorimetry uses an instrument (colorimeter) to measure absorbance, an instrumental method.
81	B	Analysis begins with collecting a representative sample of the material.
82	B	Determinate errors have a definite, correctable cause and consistent direction.
83	C	Indeterminate errors are small, uncontrollable, and scatter results randomly.
84	B	Accuracy measures the closeness of a result to the true value.
85	B	The known-concentration solution added from the burette is the titrant.
86	B	Precision is agreement among repeated measurements of the same quantity.
87	C	A faulty, uncalibrated instrument causing consistent error is an instrumental error.
88	B	An error inherent to the chosen analytical method is a method error.
89	C	An error from the analyst's personal limitation is a personal error.
90	B	Absolute error = measured - true = 12.42 - 12.50 = -0.08 g.
91	B	Percentage error = (0.50/50.00) x 100 = 1.0%.
92	C	Range is defined as the highest value minus the lowest value.
93	C	In 0.02030, digits 2, 0, 3, 0 are significant; leading zeros are not.
94	B	Standard deviation uses every measurement's deviation from the mean, unlike range.
95	B	Primary standards need high purity, stability, and a known high equivalent weight.

96	**B**	NaOH absorbs moisture and carbon dioxide from air, so its purity is uncertain.
97	**B**	End point may not exactly match equivalence point; the gap is the titration error.
98	**C**	$N_1V_1 = N_2V_2$: $N_1 \times 20 = 0.15 \times 24 = 3.6$, so $N_1 = 3.6/20 = 0.18$ N.
99	**C**	meq HCl = 0.2x40 = 8; mass Na_2CO_3 = 8x53/1000 = 0.424 g; % = 0.424/0.530x100 = 80%.
100	**B**	Consistent small error with tight repeatability means high precision, low accuracy, personal error.

ANSWER KEY + ONE-LINE WORKING — MATHEMATICS (Q101–150)

Q		Working
101	C	sin(30deg)=1/2, cos(60deg)=1/2, sum=1.
102	A	tan(45deg)=1, cos(0deg)=1, product=1.
103	D	cosec(30deg)=1/sin(30deg)=1/(1/2)=2.
104	B	sec(60deg)=1/cos(60deg)=1/(1/2)=2.
105	C	tan(60deg)=root3, tan(30deg)=1/root3, ratio=3.
106	D	sin(120deg)=sin(180deg-60deg)=sin(60deg)=root3/2, positive in QII.
107	B	cos(210deg)=cos(180deg+30deg)=-cos(30deg)=-root3/2, cos negative in QIII.
108	A	tan(315deg)=tan(360deg-45deg)=-tan(45deg)=-1, tan negative in QIV.
109	C	By identity sin^2(theta)+cos^2(theta)=1 for any theta, including 30deg.
110	A	Both sin and cos are positive only in quadrant I (ASTC rule).
111	D	sin(theta) and cos(theta) repeat their values every 360deg.
112	B	tan(theta) is undefined where cos(theta)=0, i.e., at odd multiples of 90deg.
113	C	cos(theta) lies in [-1,1], so sec(theta)=1/cos(theta) lies outside the interval -1 to 1.
114	D	In QIII only tan (and cot) are positive; sin and cos are negative.
115	A	tan positive in I and III; sin negative in III and IV; common quadrant is III.
116	B	sin(-theta)=-sin(theta) and cos(-theta)=cos(theta), so product=-sin(theta)cos(theta).
117	C	90+theta lies in QII where sin is positive; co-function change gives cos(theta).
118	D	270 is an odd multiple of 90, giving co-function cot(theta), positive in QIII.
119	A	From identity 1+tan^2(theta)=sec^2(theta), rearranged gives sec^2(theta)-tan^2(theta)=1.
120	B	From identity 1+cot^2(theta)=cosec^2(theta), rearranged gives cosec^2(theta)-cot^2(theta)=1.
121	C	sin(theta)+1/sin(theta)=2 forces sin(theta)=1, so sin^2(theta)+cosec^2(theta)=1+1=2.
122	D	Equals sec^2(theta) x (1-sin^2(theta)) = sec^2(theta) x cos^2(theta) = 1.
123	A	sec^2(theta)-tan^2(theta)=1, so (sec(theta)+tan(theta))(sec(theta)-tan(theta))=1, giving 1/p.
124	B	Simplifies to sin(theta)/(1+cos(theta)) + (1+cos(theta))/sin(theta) = 2/sin(theta) = 2cosec(theta).
125	C	sin(theta)=-4/5 (negative in QIII), tan(theta)=sin(theta)/cos(theta)=(-4/5)/(-3/5)=4/3.
126	B	cosA=4/5, sinB=5/13; sin(A+B)=sinAcosB+cosAsinB = 36/65+20/65 = 56/65.
127	C	sinA=3/5, cosB=12/13; cos(A-B)=cosAcosB+sinAsinB = 48/65+15/65 = 63/65.
128	A	sin75deg = sin(45deg+30deg) = sin45cos30+cos45sin30 = (root6+root2)/4.
129	D	tan(A+B) = (1/2+1/3)/(1-1/6) = (5/6)/(5/6) = 1.
130	B	cos2A = 2cos^2A - 1 = 2(9/25) - 1 = -7/25.
131	C	cosA=4/5; sin2A = 2sinAcosA = 2(3/5)(4/5) = 24/25.
132	A	tan2A = 2tanA/(1-tan^2A) = 1/(1-1/4) = 4/3.
133	D	tan15deg = tan(45deg-30deg) = (1-tan30)/(1+tan30) = 2-root3.
134	B	cosA=12/13, sinB=4/5; sin(A-B)=sinAcosB-cosAsinB = 15/65-48/65 = -33/65.
135	C	sin(A/2) = root((1-cosA)/2) = root((18/25)/2) = root(9/25) = 3/5.
136	A	sinC+sinD = 2 sin((C+D)/2) cos((C-D)/2), giving 2 sin3x cosx.
137	D	cosC-cosD = -2 sin((C+D)/2) sin((C-D)/2), giving -2 sin4x sinx.
138	B	2cosA cosB = cos(A+B) + cos(A-B) = cos4x + cos2x.
139	C	sin3A = 3sinA - 4sin^3A = 1 - 4/27 = 23/27.
140	A	cos3A = 4cos^3A - 3cosA = 4(1/8) - 3(1/2) = -1.
141	D	b = a sinB/sinA = 10 x (root2/2)/(1/2) = 10root2.
142	B	cosA = (b^2+c^2-a^2)/(2bc) = (36+49-25)/84 = 60/84 = 5/7.
143	C	Standard projection formula: a = b cosC + c cosB.
144	A	sinC+sinD = 2sin30deg cos20deg = 2(1/2)cos20deg = cos20deg.
145	D	General solution of sinx = sin(pi/6) is x = npi + (-1)^n (pi/6), n is an integer.

146	**B**	cosx = cos(2pi/3), so general solution is x = 2npi plusminus 2pi/3.
147	**C**	(2cosx-1)(cosx-1)=0 gives cosx=1/2 or cosx=1, so x=2npi or 2npi plusminus pi/3.
148	**A**	Numerator=2cos4x sinx, denominator=2cos4x cosx; ratio simplifies to tanx.
149	**D**	Combine over common denominator: sin(3A-A)/(sinAcosA) = sin2A/(sinAcosA) = 2.
150	**B**	c^2=a^2+b^2-2ab cosC = 8 - 2(2)(1/2) = 8-2 = 6, so c = root6.